



Assignment 4

RL

1 Problem Statement

Consider the following 5×5 world (shown in the figure).

R1	1	0	-1	R2
2	1	0	-1	-2
2	1	0	-1	-2
2	1	0	-1	-2
2	1	0	-1	-2

The agent has four actions: Up, Down, Right and Left. The transition model is:

- If the intended action is **Up**, then:
 - 70% to Up,
 - 10% to Down,
 - 10% to Left,
 - 10% to Right
- If the intended action is **Down**, then:
 - 70% to Down,
 - 10% to Up,
 - 10% to Left,
 - 10% to Right



- If the intended action is **Right**, then:
 - 70% to Right,
 - 10% to Left,
 - 10% to Down,
 - 10% to Up
- If the intended action is **Left**, then:
 - 70% to Left,
 - 10% to Right,
 - 10% to Down,
 - 10% to Up
- A collision with a wall results in no movement.

2 Requirements

1. Implement value iteration for this world for each values of the reward R_1 , R_2 below:
 - $R_1 = 100, R_2 = 110$
 - $R_1 = 10, R_2 = 100$
 - $R_1 = 1, R_2 = 10$
 - $R_1 = 10, R_2 = 15$
2. Use discounted rewards with a discount factor of 0.95.
3. Show the policy obtained in each case.
4. Explain intuitively why the values of R_1 , R_2 leads to each policy.

2.1 Bonus

Find the optimal policy for each of the previous cases using the Policy Iteration algorithm. You must start the algorithm with a randomly generated policy.



2.2 Further Notes

- You may use Java, Python, or C++ for your implementation.
- Copied assignments will be severely penalized, whether copied from online sources or from each other.
- You can work in groups of 3 only.

Good Luck